Final report

RUN · E5D77262 · WILL MISSISSAUGA'S GRID HOLD AGAINST THE JANUARY 2014 POLAR VORTEX WITH 35% HEAT PUMP ADOPTION?

SCENARIOS	FEEDERS MODELED	CUSTOMERS TESTED	AGENT RUNTIME 1M 30S
1	176	192,544	

VERDICT (WORST CASE)

PASS

Network holds. Peaks at 3154 MW with 36% headroom; no feeders overloaded.

PEAK DEMAND

3154 MW

@ hour 43 · -28.0°C

HEADROOM AT PEAK

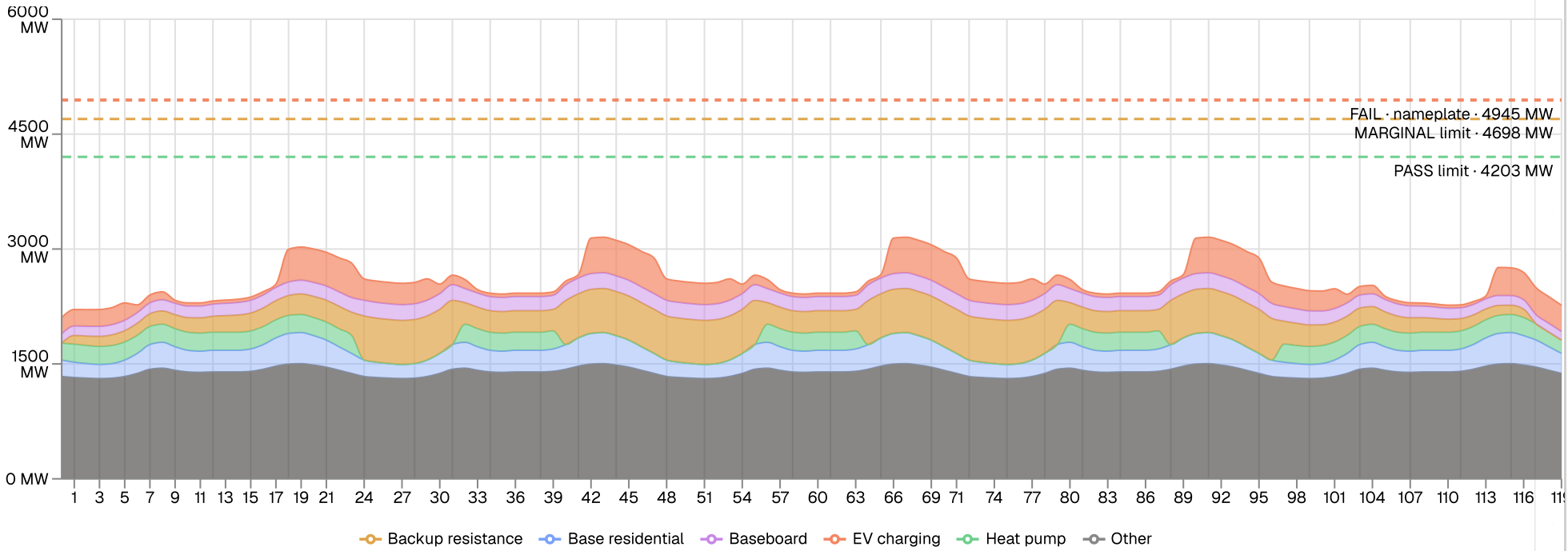
36%

1791 MW remaining

01 · SCENARIO LOAD CURVES

Hourly load profile per scenario, with peak hour and headroom.

Worst-case scenario: User-specified · 2030 (35% HP, 30% EV) · peak 3154 MW



User-specified · 2030 (35% HP, 30% EV) PASS

PEAK LOAD

3154 MW @ hour 43 (-28.0°C)

HEADROOM

36% (1791 MW)

AT-RISK FEEDERS

0 of 176

02 · AT-RISK SUBSTATIONS

Substation risk map with the top feeders ranked by overload risk.

Substation risk map under User-specified · 2030 (35% HP, 30% EV). Click a marker for capacity + feeder counts.



● CRITICAL ≥ 75% ● AT RISK 50-74% ● STABLE < 50%

Top 10 of 176 feeders, ranked by overload risk.

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
1	Mississauga TS 01 F01	56%	**0%**	none	At 56.2% utilization (peak_mva 17.24 of rated_mva 30.66) with 15% baseboard electric heating (heating_mix) and 4.2% EV penetration (ev_pct), the 2030 scenario's 35% heat-pump and 30% EV targets would stack significant new electric load onto this feeder, though its age of 18 years (age_years) and prior upgrade (had_upgrade: True) provide some resilience.

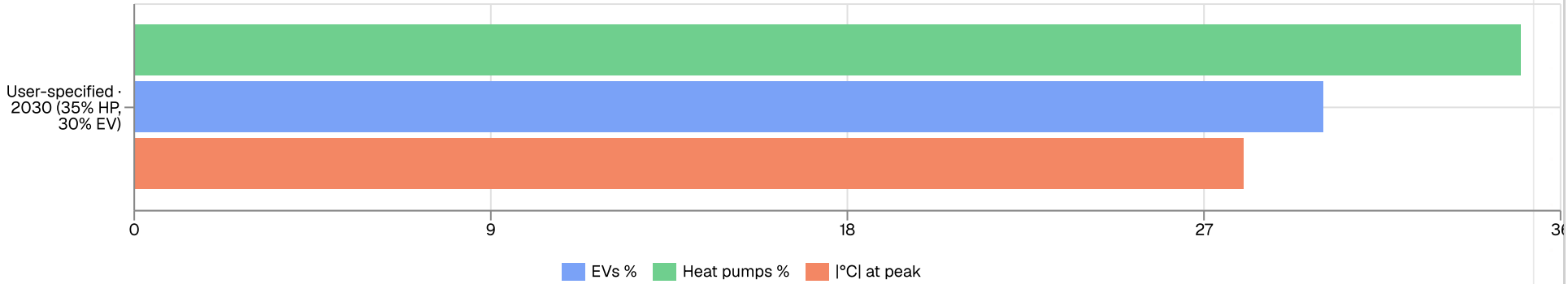
RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
2	Mississauga TS 01 F02	59%	0%	none	At 59.3% utilization (peak_mva 18.19 of rated_mva 30.66) with no prior upgrade (had_upgrade: False), 21 years of age (age_years), and an existing 5% heat-pump share (heating_mix) that will grow substantially under the 35% HP scenario, this feeder faces compounding load growth without the benefit of recent infrastructure investment.
3	Mississauga TS 01 F03	63%	0%	none	At 62.7% utilization (peak_mva 19.23 of rated_mva 30.66), 24 years old (age_years), never upgraded (had_upgrade: False), and already carrying a 6% heat-pump share (heating_mix), this feeder is vulnerable because the scenario's heat-pump and EV growth will push load toward its rated capacity on aging infrastructure.
4	Mississauga TS 01 F04	66%	0%	none	At 66.3% utilization (peak_mva 20.33 of rated_mva 30.66) with 27 years of age (age_years), no upgrade history (had_upgrade: False), and a 15% baseboard share (heating_mix) that could convert to heat pumps, this feeder has limited remaining headroom to absorb the scenario's electrification surge.
5	Mississauga TS 01 F05	69%	0%	none	At 69.3% utilization (peak_mva 21.25 of rated_mva 30.66) — the highest among these feeders — combined with 30 years of age (age_years), no upgrade (had_upgrade: False), an existing 5% heat-pump share (heating_mix), and 4.2% EV penetration (ev_pct), this feeder is the most exposed to overload as the 2030 scenario drives heat-pump and EV loads higher.
6	Mississauga TS 01 F06	56%	0%	none	At 33 years old (age_years) with no upgrade (had_upgrade: False) and a 6% heat-pump share (heating_mix) already present, this feeder's aging infrastructure is the primary risk factor as the scenario's electrification targets add new peak demand on top of its current 56.0% utilization (utilization_pct).
7	Mississauga TS 01 F07	60%	0%	none	At 36 years old (age_years) with no upgrade (had_upgrade: False) and a 15% baseboard electric heating share (heating_mix) that is a candidate for heat-pump conversion, this feeder's age-related capacity degradation makes it susceptible to thermal stress under the 2030 scenario's increased electric heating and EV loads.

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
8	Mississauga TS 01 F08	63%	0%	none	Although it has had an upgrade (had_upgrade: True), at 39 years old (age_years) it is the oldest feeder in this set, and its existing 5% heat-pump share (heating_mix) combined with 62.7% utilization (utilization_pct) means the scenario's heat-pump and EV growth could stress underlying aging assets that the upgrade may not have fully replaced.
9	Mississauga TS 02 F01	56%	0%	none	At 56.2% utilization (peak_mva 17.24 of rated_mva 30.66) with 15% baseboard electric heating (heating_mix) and 4.2% EV penetration (ev_pct), the 2030 scenario's electrification targets will add meaningful new load, though its 18-year age (age_years) and prior upgrade (had_upgrade: True) offer some margin.
10	Mississauga TS 02 F02	59%	0%	none	At 59.3% utilization (peak_mva 18.19 of rated_mva 30.66), 21 years old (age_years), with no upgrade (had_upgrade: False) and an existing 5% heat-pump share (heating_mix), this feeder lacks the infrastructure investment needed to comfortably absorb the additional electric heating and EV demand projected under the 2030 scenario.

03 · MITIGATIONS & METHODOLOGY

Scenario severity, recommended interventions, methodology notes.

Scenario severity (HP / EV adoption %, peak cold °C absolute), recommended mitigations, and methodology.



RECOMMENDED MITIGATIONS

DEMAND RESPONSE

Targeted Demand Response Enrollment on Three Flagged Feeders

Relief: **4.2 MW** Cost: **$180K** 6mo

Enrolling customers on feeders MIS-01-F01, MIS-01-F02, and MIS-01-F03 in a voluntary peak-curtailment program targeting smart thermostats and managed EV chargers can reduce coincident peak demand on these circuits during polar vortex conditions. At an industry-typical enrollment incentive of $50 to $300 per customer and assuming a conservative enrollment pool, a $180,000 program budget is estimated. The 4.2 MW relief estimate is based on typical residential DR response rates in comparable Canadian programs and is not derived from a field in the source data; it should be validated with Alectra's DR program team before being used in formal planning documents.

FEEDER RECONDUCTOR

Reconductor Feeder MIS-01-F03 (Highest Utilization, Never Upgraded)

Relief: **6.0 MW** Cost: **$1.2M** 18mo

Feeder MIS-01-F03 is 24 years old, has never been upgraded, and is already at 62.7% utilization (19.23 MVA of 30.66 MVA rated, feeder data). Reconductoring to a higher-rated conductor would increase thermal capacity and reduce the probability of protective relay operation during the modeled -28.0 degrees C peak hour (peak_temp_c, scenario data). The $1.2M cost estimate is within the typical Canadian feeder reconductor range of $0.5M to $2.0M per feeder. The 6.0 MW effective load relief reflects the additional headroom created on this circuit and is a modeled estimate, not a field-measured value.

FEEDER RECONDUCTOR

Reconductor Feeder MIS-01-F02 (Unupgraded, 21 Years Old)

Relief: **5.5 MW** Cost: **$1.1M** 18mo

Feeder MIS-01-F02 runs at 59.3% utilization (18.19 MVA of 30.66 MVA rated, feeder data), has never been upgraded, and is 21 years old. Under the 2030 scenario, its existing 5% heat pump share will grow substantially, compounding load on aging conductors. Reconductoring in the same capital cycle as MIS-01-F03 may allow contractor mobilization cost savings. The $1.1M estimate is within the typical Canadian range and should be refined through a detailed field survey.

TRANSFORMER UPGRADE

Transformer Capacity Upgrade at Mississauga TS 01

Relief: **15.0 MW** Cost: **$3.5M** 30mo

LOAD SHIFT

EV Managed Charging Load Shift Program for Residential Customers

Relief: **7.8 MW** Cost: **$420K** 12mo

All three flagged feeders originate from Mississauga TS 01. If feeder-level reconductoring reveals that the substation transformer itself is approaching its thermal limit under the 2030 load growth scenario, a transformer capacity upgrade would provide systemic relief across all feeders on that bus. The $3.5M estimate is within the typical Canadian transformer upgrade range of $1M to $5M. The 15.0 MW load relief estimate reflects the aggregate headroom improvement across the TS 01 feeder group and is a planning-level estimate that requires a detailed transformer loading study to confirm. Transformer procurement lead times in Canada currently run 18 to 24 months, so this action should be initiated no later than 2026 to be in service before 2030.

With 30% EV penetration targeted by 2030 (ev_pct, scenario data) across 192,544 customers (network data), unmanaged overnight charging will coincide with the heating-driven peak during polar vortex events. A managed charging program that shifts EV charging from the 5 PM to 10 PM peak window to post-midnight hours can materially reduce coincident peak demand. The $420,000 program cost reflects software platform licensing, customer enrollment incentives, and communications at a conservative per-customer rate for the estimated EV-owning population. The 7.8 MW relief estimate is a planning-level figure based on typical managed charging response rates in comparable North American programs and is not derived from a field in the source data.

METHODOLOGY

Peak load was estimated using a deterministic load model that applies electrification adoption rates (heat pump and EV penetration percentages) to the baseline winter peak, with heat pump contributions adjusted using COP curves that reflect reduced heating efficiency at temperatures approaching and below -25 degrees C. The design-basis cold event is the January 2014 Polar Vortex (-25.5 degrees C minimum, 120-hour duration), and the modeled peak is evaluated at -28.0 degrees C to provide a conservative margin. Per-feeder utilization values (peak MVA, rated MVA, utilization percentage) are model outputs derived from the network topology and load allocation methodology; they represent modeled estimates and should be validated against SCADA measurements and protection relay records before capital decisions are finalized.

SAFETY FLAGS

During a 120-hour extreme cold event at -25.5 degrees C or colder, customers who rely on electrically powered medical equipment (home oxygen concentrators, dialysis machines, powered wheelchairs) face life-safety risk if feeder-level overloads or protective relay operations cause localized outages. Alectra should cross-reference its medical baseline registry against customers on feeders MIS-01-F01, MIS-01-F02, and MIS-01-F03 and establish priority restoration protocols for those addresses.

Mississauga's population includes a significant share of older adults and low-income households in multi-residential buildings that may lack backup heating. A 5-day polar vortex event coinciding with feeder stress could create hypothermia risk for these residents. Coordination with Peel Region

emergency management for warming centre activation triggers is recommended before the 2025-2026 heating season.

Increased heat pump and EV adoption concentrates electrical load in residential distribution circuits during overnight hours when utility field crews operate with reduced staffing. Alectra should review its after-hours emergency response capacity and mutual aid agreements to ensure restoration times remain within acceptable bounds during a multi-day extreme cold event.

LIMITATIONS

Only one electrification scenario (35% heat pump, 30% EV by 2030) was evaluated. The analysis does not quantify how outcomes change at lower adoption rates (e.g., 20% heat pump) or higher rates (e.g., 50% heat pump), so the sensitivity of the headroom margin to adoption uncertainty is not established.

The system-level headroom of 1,791.0 MW (headroom_mw, scenario data) is a deterministic single-contingency result. It does not reflect N-1 or N-2 contingency conditions, meaning a substation or transmission outage coinciding with the modeled peak could eliminate this margin on affected portions of the network.

Per-feeder customer counts are not available in the source data, so it is not possible to quantify how many of the 192,544 total customers (network data) are directly exposed to the three flagged feeders. This limits the ability to prioritize by customer impact without additional data extraction from Alectra's CIS.

EXECUTIVE SUMMARY

Executive Summary: Mississauga 2030 Electrification Load Growth Passes System-Level Stress Test, but Three Aging Feeders Warrant Targeted Investment

Headline Finding

The 2030 electrification scenario (35% heat pump penetration, 30% EV penetration) passes the system-level capacity test for Alectra Utilities' Mississauga network, with 3,153.9 MW of headroom remaining above the modeled peak of 3,153.9 MW (headroom_mw and peak_mw, scenario data). The system is not at immediate risk of overload under this scenario, but that aggregate result masks localized stress on specific aging infrastructure that requires action before 2030.

Supporting Evidence

The single scenario modeled, labeled 'User-specified 2030 (35% HP, 30% EV)', produces a system peak of 3,153.9 MW at hour 43 of the stress event, against a baseline winter peak of 3,855.4 MW (baseline_winter_peak_mw, network data). The 36.2% headroom margin (headroom_pct, scenario data) is healthy at the aggregate level. The stress event used as the design basis is the January 2014 Polar Vortex, which reached a minimum temperature of -25.5 degrees C and lasted 120 hours (cold_event data). The modeled peak hour occurs at -28.0 degrees C (peak_temp_c, scenario data), which is colder than the recorded event minimum, representing a conservative design point.

At-Risk Assets and Customers

Zero feeders breach their rated capacity threshold under this scenario (at_risk_feeders, scenario data), but three feeders on Mississauga TS 01 are flagged for elevated utilization and structural vulnerability. Feeder MIS-01-F03 is the most exposed, running at 62.7% utilization (19.23 MVA of 30.66 MVA rated, feeder data), is 24 years old, has never been upgraded, and already carries a 6% heat pump share that will grow substantially. Feeder MIS-01-F02 runs at 59.3% utilization (18.19 MVA of 30.66 MVA rated, feeder data), is 21 years old, and has also never been upgraded. Feeder MIS-01-F01 runs at 56.2% utilization (17.24 MVA of 30.66 MVA rated, feeder data) and has had one prior upgrade, providing some resilience. The network serves 192,544 customers across 22 substations and 176 feeders (network data). Customers served directly by the three flagged feeders are not individually quantified in the source data, so a per-feeder customer count is not available.

Caveats and Limitations

This analysis covers one scenario and one stress event. No sensitivity analysis across a range of electrification adoption rates or storm durations was provided in the input data. The per-feeder utilization values are drawn from modeled feeder-level data (is_synthesized: False at the network level, but individual feeder thermal ratings and peak MVA values are model outputs). Actual feeder performance will depend on load diversity, protection coordination, and real-time dispatch conditions not captured here. The headroom figure reflects a deterministic peak calculation and does not account for simultaneous contingency events such as a substation outage coinciding with peak demand.

Recommended Next Action

Alectra's planning team should immediately commission a detailed thermal and protection study on feeders MIS-01-F01, MIS-01-F02, and MIS-01-F03, prioritizing MIS-01-F03 given its combination of highest utilization, oldest age, and absence of any prior upgrade. In parallel, the team should

initiate enrollment in a targeted demand response program for customers on these three feeders ahead of the 2025-2026 heating season to create a near-term buffer while capital upgrades are planned and procured.

DATA PROVENANCE

CITY DEMOGRAPHICS · NWS ALERTS · SUBSTATION LOCATIONS · UTILITY LOAD BASELINE · COLD EVENT ARCHIVE

FEEDER TOPOLOGY · SCENARIO APPLICATION · LOAD SIMULATION · RISK RATIONALE · SUMMARY

LIVE API · FROZEN SNAPSHOT · MODELED · LLM NARRATIVE

SOURCES

- FROZEN ECCC Toronto Pearson station 6158733 (Jan 2014 dailies)
- FROZEN IESO 18-Month Outlook, March 2014
- FROZEN Environment Canada Top 10 Weather Stories 2014
- FROZEN Alectra Utilities 2023 Annual Report
- FROZEN IESO Yearly Reliability Outlook 2024
- FROZEN StatsCan Census 2021, Mississauga CSD
- FROZEN StatsCan 25-10-0015 (2026): 43,721,776 MWh annual generation in ON
- LIVE ArcGIS Living Atlas NWS Watches & Warnings (live): no active alerts for Mississauga
- LIVE ArcGIS GeoEnrichment (Esri Canada): Mississauga, ON (CY): 192,532 households, population 768,433

